



Innovate with Ballerina 2025

Awareness Session

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A decorative graphic on the left side of the slide, consisting of several overlapping, concentric, teardrop-shaped loops in shades of green and yellow.

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About Ballerina



Ballerina Swan Lake

- Fully open-source programming language, developed by WSO2
- Cloud-native programming language optimized for integration
- Data-oriented and network aware
- A feature-rich, complete platform





```
import ballerina/http;
```

```
service on new http:Listener(8080) {
```

```
    resource function get persons() returns Person[]|http:NotFound {
```

```
        // Implementation
```

```
    }
```

```
    resource function get persons/[int id]() returns Person|http:NotFound {
```

```
        // Implementation
```

```
    }
```

```
    resource function put person/[int id](Person person)
```

```
        returns Person|http:BadRequest|http:NotFound {
```

```
            // Implementation
```

```
    }
```

```
}
```



```
import ballerina/http;

service on new http:Listener(8080) {
    resource function get persons() returns Person[]|http:NotFound {
        return persons.toArray();
    }

    resource function get persons/[int id]() returns Person|http:NotFound {
        if persons.hasKey(person.id) {
            return persons.get(id);
        }
        return http:NOT_FOUND;
    }
}
```

```
type Tutor record {  
  readonly int id;  
  string name;  
  string department;  
};  
  
type Student record {  
  readonly int id;  
  string name;  
  record {  
    string code;  
    int credits;  
  }[] courses;  
};  
  
type Person record {  
  readonly int id;  
  string name;  
};
```

```
table<Person> key(id) persons = table [];  
  
Student joy = {  
  id: 1001,  
  name: "Joy Smith",  
  courses: [  
    {code: "CS101", credits: 3},  
    {code: "CS102", credits: 4}  
  ],  
};  
persons.add(joy); // Valid  
  
Tutor mary = {  
  id: 7001,  
  name: "Mary Williams",  
  department: "Computer Science"  
};  
persons.add(mary); // Valid
```

Ballerina is graphical



The image displays the Ballerina IDE interface, showing the source code of a program and its corresponding graphical flow diagram.

Source Code (automation.bal):

```
1 import ballerina/log;
2 import ballerina/googleapis.gmail;
3
4 public function main() returns error? {
5     do {
6         log:println("Sending email to: " + receiverEmail);
7
8         // construct and send the email message
9         gmail:Message gmailMessage = check gmailClient->users/[gmailUserId]/messages/send.post({
10             'from: senderEmail,
11             to: [receiverEmail,
12             subject: "Morning Update",
13             bodyInText: "Hello! This is your daily update from '" + gmailUserId + "'. " +
14                 "Have a great day!"
15         });
16
17         log:println("Email sent successfully: Email ID = " + gmailMessage.id);
18     } on fail error e {
19         log:printlnError("Error: ", 'error = e);
20         return e;
21     }
22 }
23
```

Graphical Flow Diagram:

The diagram, titled "Automation", illustrates the execution flow of the code:

- Start** node leads to a function block `log:println` with arguments `"Sending email to: " + receiverEmail`.
- A connector node leads to a comment box: `construct and send the email...`.
- The flow continues to a `gmail:post` block with argument `gmailMessage`. A `gmailClient` icon is shown as a resource.
- Next is another function block `log:println` with arguments `"Email sent successfully: Email ID = " +` and `gmailMessage.id`.
- The flow ends at an **Error Handler** block.

The status bar at the bottom indicates: `Ln 14, Col 14 Spaces: 4 UTF-8 LF () Ballerina 2201.12.3 (Swan Lake Update 12)`.



Data transformation



The screenshot displays the Ballerina IDE interface. On the left, the 'automation.bal' file is open, showing a Ballerina program that defines records for Person, Course, and Student, and includes logic for interacting with a Google Sheets API. On the right, the 'Ballerina Visualizer' is open, showing a 'GraphQL Diagram / Data Mapper' for the 'enrollPerson' operation. The diagram maps input fields from a 'person' record to a 'Student' record and a 'courses' array to a 'totalCredits' field. The status bar at the bottom indicates the current position in the file (Ln 55, Col 1) and the Ballerina version (2201.12.3).

```
1 import ballerina/http;
2 import ballerina/googleapis/sheets;
3
4 type Person record {
5     readonly int id;
6     string firstName;
7     string lastName;
8     int age;
9     string country;
10 };
11
12 type Course record {
13     string id;
14     string name;
15     int credits;
16 };
17
18 type Student record {
19     string id;
20     string fullName;
21     int age;
22     record {
23         string title;
24         int credits;
25     }[] courses;
26     int totalCredits;
27     string visaType;
28 };
29
30 configurable int port = 8080;
31 configurable string sheetsAccessToken = ?;
32 configurable string spreadsheetId = ?;
33 configurable string sheetName = "enrollments";
34
35 final sheets:Client gsheetsClient = check new ({auth: {token: sheetsAccessToken}});
36
37 const D_TIER_4_VISA = "D tier-4";
38
39 table<Person> key(id) persons = table [
40     {
41         id: 1001,
42         firstName: "John",
43         lastName: "Doe",
44         age: 25,
45         country: "LK"
46     },
47     ...
48 ];
```

Data Mapper enrollPerson

person: Person

- id: int
- firstName: string
- lastName: string
- age: int
- country: string

Student

- id*: string
- fullName*: string
- age*: int
- courses*: record[]
- totalCredits*: int
- visaType*: string

courses: Course[]

Module Variables

- D_TIER_4_VISA: string
- totalCredits: ?[] course) returns int

Ln 55, Col 1 Spaces: 4 UTF-8 LF () Ballerina 2201.12.3 (Swan Lake Update 12)

Fix using Ballerina Copilot



agents.bal M X

← → ai-samples

BI Copilot X

personal_ai_assistant_agent > agents.bal > readEmails

```
1 import ballerina/os;
2 import ballerina/ai.agent;
3 import ballerina/googleapis.calendar;
4 import ballerina/googleapis.gmail;
5
6 @agent:Tool
7 @display {
8   label: "Read Emails",
9   iconPath: "https://bcentral-packageicons.azureedge.net/images/ballerina_googleapis.gmail";
10 }
11
12 Visualize
13 isolated function readEmails() returns gmail:Message[]|error {
14   gmail:ListMessagesResponse messageList = check gmailClient->/users/me/messages(q = "Label
15   gmail:Message[] messages = messageList.messages ? : [];
16   gmail:Message[] completeMessages = [];
17   foreach gmail:Message message in messages {
18     gmail:Message completeMsg = check gmailClient->/users/me/messages/{message.id} (forma
19     completeMessages.push(completeMsg);
20   }
21   return completeMessages;
22 }
23
24 // Personal AI Assistant
25
26 @agent:Tool
27 @display {
28   label: "Send Email",
29   iconPath: "https://bcentral-packageicons.azureedge.net/images/ballerina_googleapis.gmail";
30 }
31
32 Visualize
33 isolated function sendEmail(string[] to, string subject, string body) returns gmail:Message|
34   gmail:MessageRequest requestMessage = {to, subject, bodyInText: body};
35   gmail:Message message = check gmailClient->/users/me/messages/send.post(requestMessage);
36   return gmailClient->/users/me/messages/{message.id}/modify.post({removeLabelIds: ["UNREA
37 }
38
39 @agent:Tool
40 @display {
41   label: "Get Calander Events",
42   iconPath: "https://bcentral-packageicons.azureedge.net/images/ballerina_googleapis.cale
43 }
44
45 Visualize
46 isolated function getCalanderEvents() returns stream<calendar:Event, error?>|error {
47   return calendarClient->getEvents(userEmail);
48 }
49
50 @agent:Tool
```

Welcome to BI Copilot

Integrate better with your AI pair.

BI Copilot uses AI to assist with integration. Please review all suggested content before adding it to your integration. By signing in, you agree to our [Terms of Use](#).

Login to BI Copilot

main*

0 2

Ballerina 2201.12.3 (Swan Lake Update 12)



Ballerina for AI

```
import ballerina/ai.agent;

final agent:OpenAiModel openAiModel = check new (openAiApiKey, agent:GPT_40);

final agent:Agent personalAiAssistantAgent = check new (systemPrompt = {
    role: "Personal AI Assistant",
    instructions: string `You are Nova, an intelligent personal AI assistant designed to help '${userName}'
stay organized and efficient.
```

Your primary responsibilities include:

- Email Assistance: Reading, summarizing, composing, and sending emails.
- Privacy & Security: Handling user data responsibly, ensuring sensitive information is kept confidential.

Guidelines:

- Respond in a natural, friendly, and professional tone.
- Always confirm before making changes to the user's calendar or sending emails.
- Provide concise summaries when retrieving information unless the user requests details.`

```
}, model = openAiModel, tools = [readEmails, sendEmail]);
```



Sample Use Case

Sample use case



Mobile application

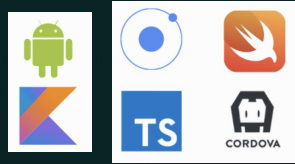


Web application



Business Logic

Sample use case



Mobile application

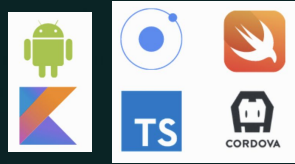


Web application



Business Logic

Sample use case



Mobile application

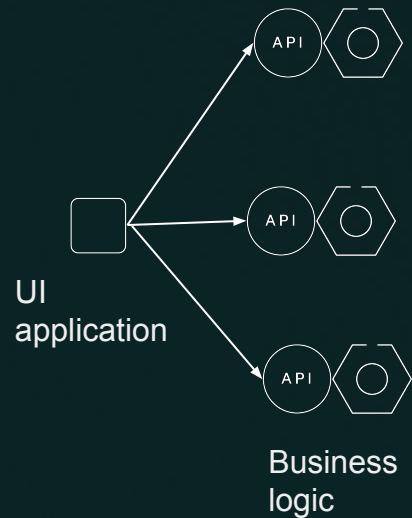
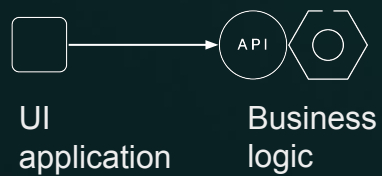


Web application

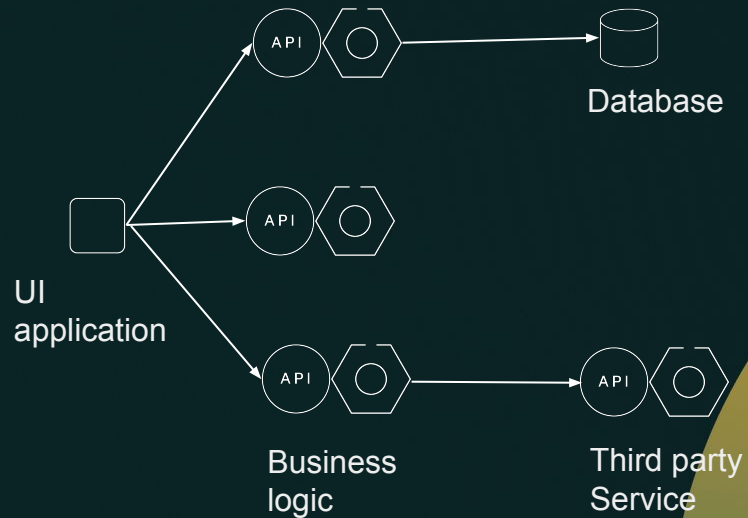
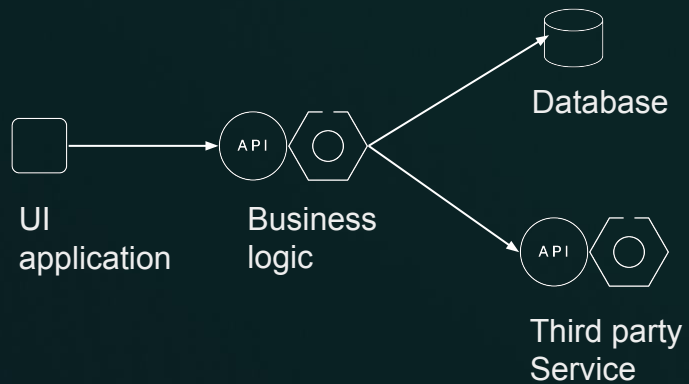


Business Logic

Sample use case



Sample use case





Build with Ballerina

API Integration

A weather assistant that sends travel suggestions based on local weather conditions.

Microservices Orchestration

A distributed healthcare system that automates scheduling, reminders, and patient data processing.

Event-Driven Integration

Event-driven emergency alert system that integrates sensors and communication platforms to provide notifications in the event of disasters.

Generative AI Applications

Build smart content generators, personalized assistants, or real-time translators that respond to user inputs and context.





What you need to do

Round 1 - Virtual

- Implement your hackathon idea with Ballerina
- Upload the source code to GitHub as a public open-source repository.
- Create a 5 to 10-minute demonstration video and upload it to YouTube.
- Submit the GitHub repository URL and YouTube video URL
- Feel free to share a social media post or article highlighting the potential impact of your work and your team's experience with the competition.



What you need to do

Round 2 - Physical

- Complete the round 2 task (revealed on the day of the event)
- Present your project to the judges





Evaluation Criteria

- **Innovation**

Originality and creativity of the idea

- **Impact**

The potential business/social value and the practicality

- **Ballerina usage**

Utilizing the features and capabilities of the Ballerina language

- **Presentation**

Clear articulation of the problem and solution



Resources and Support



Learning material

- Download Ballerina Swan Lake - <https://ballerina.io/downloads/>
- Ballerina documentation and tutorials
 - [Learn guide](#)
 - [Ballerina by example](#)
 - [Youtube training series](#)
 - [Ballerina use cases \(Integrations, AI, EDA, etc\)](#)
- Innovate with Ballerina 2024 Workshops
 - [Awareness session](#)
 - [Workshop 1 - Backend development basics](#)
 - [Workshop 2 - Advanced Ballerina Concepts](#)





Join the Ballerina community



Discord

[ballerinalang](https://discord.com/invite/ballerinalang)



stackoverflow

[WSO2 Collective](https://stackoverflow.com/questions/tagged/ballerina)



[@ballerinalang](https://twitter.com/ballerinalang)



GitHub

[ballerina-lang](https://github.com/ballerina-lang)





Thank you